Age-Invariant Identity Matching from Naturally-Anchored Social-Media Posts: A Naturally-Supervised Longitudinal Signal

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***Abstract*—Cross-age face verification—deciding whether two photographs taken years apart depict the same person—is challenging: age substantially alters appearance, and models tend to exploit age as a discriminative shortcut. We study a *data-centric* question: can the supervision needed for age invariance be obtained without a manually labeled longitudinal dataset? We show that multi-photo "then/now" social-media posts depicting one person at different ages provide naturally anchored positive identity pairs, and that fine-tuning a weak recognizer on them induces *transferable* age invariance. Our supervision is manual-label-free (no human identity/age annotation), though the pipeline uses a large-language-model caption parser and embedding clustering; we therefore call it *naturally supervised* and audit those components. On the external cross-age benchmark FG-NET, fine-tuning a deliberately weak trainable backbone raises large-gap ($\geq$25-year) ROC-AUC from 0.736 to 0.848 ($+$0.112 $\pm$ 0.001 over three seeds) at a cost of only $-$0.019 LFW accuracy; on AgeDB-30 and CALFW the model does not collapse (a slight decrease and essentially no change, respectively), so the gain is concentrated in the large-gap regime. The effect (a) exceeds synthetic aging even with a real generative re-aging model; (b) reproduces across three loss families, two architectures, two independent same-platform source communities, and two training objectives—our simple pair-contrastive objective matches the canonical ArcFace-margin classification objective on the key external metric (0.848 vs. 0.851). The result is thus a *targeted* large-gap robustness, not a uniform verification gain; we make the bounded claim that, within this protocol and across the objectives tested, the naturally-supervised data contributes more to the large-gap gain than the choice among loss families. The gain is mechanistically tied to removing an age shortcut: frozen models' accuracy degrades toward chance under age-matched negatives, and our pairs restore it. The method is data-efficient ($\approx$96% of the gain at 10% of identities). In an *apparent* gender/age audit, no stratum degrades and the largest, statistically significant gain is for the youngest (0–17) group where the frozen baseline is weakest. We additionally provide a calibrated $P(\text{same} \mid \text{cosine}, \text{age-gap})$ and a curation pipeline whose chief finding is that about half of the raw "positive" pairs were near-duplicate frames inflating internal metrics.**

***Index Terms*—Cross-age face recognition, age-invariant representation, naturally/weakly-supervised identity, dataset curation, shortcut learning, fairness, calibration.**

I. Introduction

CROSS-AGE face verification arises in archival retrieval, account recovery, and authorized search for missing persons. The challenge lies not in discriminating among arbitrary individuals but in distinguishing *the same person across age* from *different people of similar age and appearance*. Annotating longitudinal identity pairs (one individual across many years) is costly, and such data remain scarce, which constrains supervised approaches; recent work continues to report weak cross-dataset generalization and a shortage of high-quality longitudinal data.

Hypothesis. A multi-photo "then/now" post showing one person at different ages yields $\binom{n}{2}$ positive identity pairs without manual annotation; a caption stating ages adds an age anchor. This is a scalable, naturally-supervised signal of cross-age identity, and fine-tuning a weak recognizer on it should transfer to external benchmarks.

Contributions (bounded). (1) A naturally-supervised (manual-label-free) longitudinal signal mined from social posts. (2) A rigorous *attribution protocol* (one weak trainable backbone, frozen vs. fine-tuned, external evaluation) isolating the contribution of *data* from network capacity. (3) Mutually reinforcing controls (loss family, architecture, source community, training objective, real-vs-synthetic, shortcut diagnosis) supporting the bounded claim that, within this protocol and across the objectives tested, the data contributes more to the large-gap gain than the choice among loss families. (4) A curation pipeline with an audit of its automatic components, and the finding that naive positive counts are inflated about $2\times$ by near-duplicate frames. (5) Applied calibration and an apparent-demographic fairness audit with confidence intervals.

We emphasize that the novelty is data-centric rather than architectural—a new *origin of supervision*—and should be evaluated accordingly.

II. Related Work and Positioning

Three lines of prior work are most relevant. *Discriminative age-invariant FR on labeled data:* OE-CNN (orthogonal embedding decomposition) [1], DAL (decorrelated adversarial learning) [2], and MTLFace (multi-task recognition, age synthesis and attention decomposition) [3] achieve strong accuracy but are trained on datasets with *manual identity and age labels* (CACD [4], MORPH [5], AgeDB [6], FG-NET [7]). *Cross-age contrastive with synthetic samples:* CACon [8] adds a synthesized cross-age sample and a triplet loss; OrdCon [9] uses order-enhanced contrastive learning for generalized age features. *Synthetic aging for robustness:* systematic studies show synthetic aging helps only partially [10]. Generic self-supervised FR (SimCLR-style augmentation [11]) lacks identity

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Code, configurations and the de-identified benchmark protocol are available at `https://github.com/deneal123/ChildVSAdult`.

anchors across time. We instead mine the structure "one post = one person at different ages" as a label-free positive-pair generator. Table I positions us against these methods.

We report on the canonical cross-age benchmarks of these methods (AgeDB-30, CALFW [12], FG-NET; CALFW was designed as a harder, age-gapped LFW [13]). The goal is *not* to surpass the absolute state of the art (we use a weak backbone for clean attribution) but to quantify the magnitude and transferability of the gain from a naturally-supervised source; Sec. V-F trains the canonical SOTA objective on our data directly.

III. Data: Naturally-Anchored Longitudinal Pairs

Source. Two public "then/now" multi-photo communities on VK (one platform, Russian-language context). Faces are detected and aligned with a 5-point ArcFace template (InsightFace RetinaFace [14], `buffalo_l`). A *usable* face passes detection-score, minimum-size, blur, and single-target checks (Sec. IX). The raw mined corpus is summarized in Table II (left).

A. Age Extraction

Captions are parsed by regular expressions (Russian patterns plus the slash format "6/18/21", one age per photo) and by a large language model (GigaChat). An LLM audit of all 33,697 multi-photo captions (Table III) found the LLM substantially more accurate than regex—it avoids reading calendar years or gaps as ages and recovers second ages—disagreeing on 27.2% of posts, overwhelmingly LLM-favoring.

B. Persons and Leakage-Safe Split

A naive "one post = one identity" undercounts: a person recurs across posts. We cluster post-groups into persons by embedding similarity (cosine ≥ 0.85), giving 21,848 persons from 27,487 post-groups ($\approx$20% recurrence), and split *by person* (not by post). Manual inspection of high-cosine pairs showed cosine does not separate look-alikes from same-person in the 0.55–0.8 band, so the merge threshold is deliberately conservative (0.85).

C. Auditing the Automatic Supervision

Because the signal is *naturally* (not human-) supervised, the automatic components are potential hidden label noise and are audited: LLM-vs-regex age agreement (Table III); manual inspection of high-cosine pair identity; and within-person gender consistency as an inexpensive noise detector (2,919 multi-photo groups with consistency $<$0.6 flag probable multi-person). A dedicated human-audited subset (age-extraction accuracy, group-integrity accuracy, error matrix, human–LLM κ) accompanies the release.

D. Curation Pipeline and Its Largest Finding

Three steps: (i) *apparent gender/age metadata* (estimator) per face, aggregated per person—the corpus is 76.7% apparent-female / 23.3% apparent-male; (ii) *LLM group-integrity validation* (Table IV)—person-clustering had already separated most multi-person posts, so only 421 person-groups are flagged noisy; (iii) *near-duplicate dedup* within each person (cosine $\geq$ 0.97), removing 8,190 faces ($\approx$17%).

Key finding. Because the number of positives is quadratic in the number of faces per group, removing near-duplicates reduced the positive count from 65,897 to 31,865 (-52%): **approximately half of the raw "positives" were duplicate frames** (a single shoot, burst, or repost) that had inflated the internal metrics. After curation, the frozen baseline on the internal test decreases (our.overall 0.856 $\rightarrow$ 0.836; our.25+ 0.705 $\rightarrow$ 0.640), exposing a harder and more representative benchmark. The external benchmarks are unaffected. This constitutes a transferable methodological observation for mining longitudinal pairs from social media.

Threshold robustness. The -52% reduction is not an artifact of the 0.97 cutoff. Re-running the near-duplicate dedup on the pre-curation groups at cosine thresholds 0.93–0.99 (Table V) leaves 31.3k–32.4k positives across 0.93–0.97 (a stable $\approx$52% reduction) and degrades gracefully only at the very strict 0.99 (-41%): the near-duplicates are dominated by near-identical frames (cos $\approx$ 1), so the finding does not hinge on the exact cutoff.

E. Dataset Datasheet (Summary)

Provenance, rights and governance fields are stored per item (source, origin URL, collection date, license/consent status, allowed use, retention policy, deletion status, review status). Collection: public VK walls. Intended use: research on consent-based cross-age matching. Out of scope: mass identification, surveillance, de-anonymization. Minors: the child$\leftrightarrow$adult regime is in scope only under an explicit legal/ethical framework; per-item review status is tracked and deletion-on-request is supported. A full datasheet [15] accompanies the release.

IV. Method, Protocol and Statistics

Attribution protocol. Comparing an adapter on top of a *frozen* strong ArcFace against frozen ArcFace is invalid (the adapter merely deforms near-perfect embeddings). Instead we use *one weak trainable backbone* (FaceNet InceptionResnetV1 [16], CASIA-WebFace) and compare (a) frozen $\rightarrow$ benchmark metric vs. (b) soft fine-tuning of the head on our pairs $\rightarrow$ benchmark metric. Training uses a margin contrastive loss on aligned crops. *Evaluation is on external benchmarks* (LFW, AgeDB-30, CALFW, FG-NET) for a clean attribution of transfer; the internal test (`our.*`) is a secondary in-domain diagnostic. Backbone strength is varied (Sec. V-C) to map where the data helps. The design trades absolute performance for *causal attribution*.

Backbones. FaceNet (weak); ArcFace iResNet-50 [17] (CASIA-FaceV5; weak, different architecture); AdaFace IR-50/IR-101 [18] (WebFace; strong; a clean depth control); ArcFace iResNet-100 (strong).

Benchmarks and metrics. LFW (10-fold accuracy), AgeDB-30 and CALFW (aligned pairs; ROC-AUC and 10-fold accuracy), FG-NET (ROC-AUC overall and on the large-gap $\geq$25-year subset). We deliberately do not rest the headline on

TABLE I
POSITIONING VS. PRIOR WORK.

Work	Supervision	Mechanism	Our difference
OE-CNN (2018) [1]	labeled id+age	orthogonal feature decomposition	label-free data source, not decomposition
DAL (2019) [2]	labeled id+age	adversarial id/age factorization	mined pairs + simpler objective
MTLFace (2021) [3]	labeled id+age	recognition + age synthesis (multi-task)	weaker pipeline; novel supervision origin
CACon (2023) [8]	semi-sup. + synthetic	contrastive on synthesized cross-age	real mined pairs > synthetic aging (Sec. V-B)
Synthetic Ageing (2024) [10]	synthetic	train on aged images	we make *real* mining the thesis
OrdCon (2025) [9]	age-ordered contrastive	order-enhanced features	mine natural pairs, no explicit age order

TABLE II
DATASET SCALE (RAW MINED) AND AFTER CURATION (SEC. III-D).

Stage	Posts	Photos	Usable faces	Persons (groups)	Positive pairs
Raw mined (2 communities)	44,609	84,147	49,606	21,848	65,897
After curation	—	—	**40,586**	**21,427**	**31,865**

TABLE III
AGE-EXTRACTION AUDIT: LLM VS. REGEX (33,697 CAPTIONED MULTI-PHOTO POSTS).

Category	n	Share
Agree (same ages or both empty)	24,531	72.8%
LLM filled (regex missed)	3,656	10.8%
Age mismatch	4,658	13.8%
LLM missed (regex found)	852	2.5%

TABLE IV
LLM GROUP-INTEGRITY CLASSIFICATION (33,697 POSTS).

Category	n	Share
Single (one person across ages)	31,275	92.8%
Multi-person (different people)	2,034	6.0%
Unknown	336	1.0%
Meme / collage	52	0.2%

TABLE V
DEDUP-THRESHOLD SENSITIVITY: NEAR-DUPLICATE REMOVAL ALONE, ON THE PRE-CURATION GROUPS (THE SEPARATE GROUP-INTEGRITY PRUNE REMOVES A FURTHER ≈500 POSITIVES TO REACH THE HEADLINE 31,865). STABLE ACROSS 0.93–0.97.

Dedup cos	Near-dup faces	Positives	Reduction
0.93	8,653	31,304	−52.5%
0.95	8,564	31,611	−52.0%
0.97 (used)	8,304	32,360	−50.9%
0.99	6,274	38,565	−41.5%

FG-NET alone (small, old, partial detector coverage): AgeDB-30 and CALFW show that the large-gap gain comes without a catastrophic collapse on standard cross-age benchmarks (AgeDB-30 decreases slightly, CALFW is nearly unchanged), confirming that the effect is concentrated on the FG-NET large-gap subset rather than uniform across benchmarks. We distinguish *threshold-free* metrics (ROC-AUC) from *threshold-based* ones (accuracy, TAR@FAR).

Statistics. Our *primary endpoint* is the FG-NET large-gap ($\geq$25-year) ROC-AUC, with the null hypothesis that fine-tuning on the mined pairs does not improve it over the frozen baseline (gain ≤ 0); AgeDB-30, CALFW and the internal cross-age test are secondary endpoints, and easy LFW accuracy is monitored as a forgetting control. The headline gain is mean $\pm$ std over three seeds (seed variance captures training stochasticity, not full evaluation uncertainty). The fairness audit uses a paired percentile bootstrap (1000 resamples, resampling pairs and recomputing both models on the same resample) for the *gain*, accounting for frozen/tuned correlation. Table VII reports bootstrap CIs, EER and TAR@FAR for the key benchmark comparisons (the internal test additionally uses an identity-level, person-resampling bootstrap), and a Holm correction is applied across the stratified fairness tests.

Curated dataset. All headline experiments are on the curated data of Sec. III-D: 21,427 person groups, 40,586 faces, 31,865 positives; split train 22,179 / val 4,579 / test 5,107 with balanced negatives per split.

V. RESULTS

A. Main Effect and Self-Supervised Ablation

The weak FaceNet, softly fine-tuned on our pairs, raises FG-NET large-gap ROC-AUC **0.736** $\rightarrow$ **0.848** (+0.112 $\pm$ 0.001), while easy LFW falls only −0.019 and AgeDB-30/CALFW move negligibly (Table VI). An ablation isolates the source: same-post pairs use no manual labels—neither identity nor age—and deliver almost all of the gain; adding explicit age anchors contributes only +0.009 on the 25+ bucket (pre-clean ablation). The method is therefore essentially *self-supervised by identity*, with age supervision contributing only marginally.

The gain is statistically robust (std $\leq$ 0.003) and survives curation; the internal cross-age gain even *grows* on the harder curated test (our.25+ +0.195) because curation lowers the frozen baseline to 0.640 while +pairs recovers to 0.835.

Bootstrap confidence intervals (seed 42) confirm the primary endpoint: FG-NET large-gap rises from 0.736 [0.70, 0.78] to

TABLE VI
MAIN RESULT: WEAK FACENET FROZEN VS. +PAIRS, THREE SEEDS, CURATED DATA. PRE-CLEAN GAIN SHOWN FOR REFERENCE.

Metric	Frozen	+pairs (mean ± std)	Gain	(Pre-clean gain)
FG-NET large-gap	0.7361	0.8484 ± 0.0007	+**0.1124**	+0.1215
FG-NET ROC	0.8955	0.9230 ± 0.0006	+0.0275	+0.0277
our.25+ (internal)	0.6403	0.8348 ± 0.0027	+**0.1946**	+0.1677
our.overall (internal)	0.8363	0.9163 ± 0.0009	+0.0801	+0.0764
LFW accuracy	0.9688	0.9503 ± 0.0031	−0.0185	−0.0283
AgeDB-30 ROC	0.9530	0.9462 ± 0.0013	−0.0068	−0.0155
CALFW ROC	0.9482	0.9489 ± 0.0014	+0.0007	−0.0030

0.848 [0.82, 0.88] with *non-overlapping* 95% CIs; the equal-error rate nearly halves (0.335→0.219) and TAR@FAR=1% nearly doubles (0.21→0.39). On the internal 25+ test the gain holds under the more conservative *identity-level* bootstrap (resampling persons, not pairs): +pairs 0.838 [0.80, 0.88] vs. frozen 0.640 [0.59, 0.69], an interval wider than the pair-level one as expected when pairs share identities. Operating points and intervals for all benchmarks are in Table VII.

B. Real Pairs Outperform Synthetic Aging

Replacing real positives with synthetically aged versions—a domain proxy and a real re-aging network (FRAN [19])—not only fails to help but *degrades* performance on the curated data: FRAN reaches FG-NET large-gap 0.727 (*below* frozen 0.736) and our.25+ falls to 0.549 (below frozen 0.640), whereas real pairs yield +0.112 / +0.198 (Table VIII). Identity-preserving aging reproduces texture artifacts rather than genuine longitudinal variation (pose, camera, era). A native-resolution control rules out a crop-resolution artifact: re-cropping a train subset at 512 px directly from the raw images (vs. our stored 112-px crops, which FRAN upscales internally) and re-aging leaves the result unchanged—FG-NET large-gap 0.771 vs. 0.778 and our.25+ 0.671 vs. 0.670 on this 499-face subset—so synthetic aging fails to match real pairs regardless of source resolution.

C. Headroom Curve and the Strong-Backbone Regime

The gain reproduces on FaceNet, ArcFace iResNet, and AdaFace IR-50/IR-101 (three loss families, two architectures) and *monotonically decreases with frozen strength* (Table IX, Fig. 1); a clean depth control (AdaFace IR-50 → IR-101) confirms it. Strong saturated backbones do not improve and forget slightly under fine-tuning; a gentler learning rate (1e-5) approximately halves the forgetting but does not convert it into a gain. The contribution is therefore specific to weak and medium backbones with cross-age headroom—a limitation we state explicitly (Sec. VII).

D. Data-Scaling Law

Varying the fraction of training identities (val/test fixed), the gain saturates early: **≈96% of the full FG-NET large-gap gain is reached with only 10% of identities** (≈1,500), reaching a plateau from 25% (Table X, Fig. 2; each point is mean ± std over three seeds); the slight dip at full data (0.50→1.00) lies within seed variance—the curve is flat, not monotonic, beyond ≈10%. The signal is thus data-efficient; further gains require *harder* data (larger gaps, hard positives) rather than merely a greater volume.

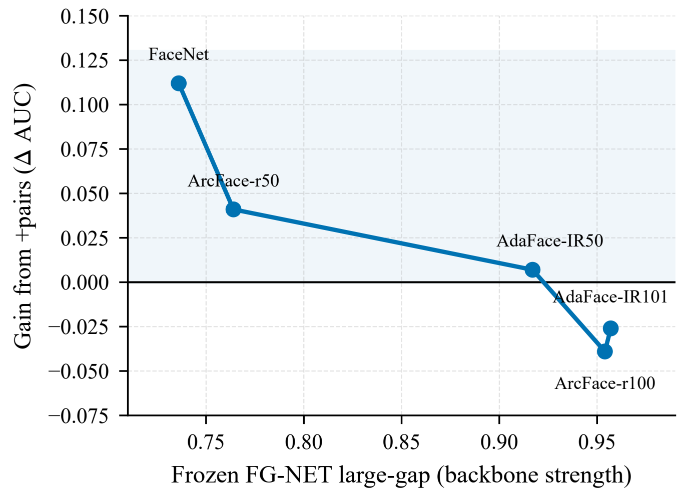

Fig. 1. Headroom curve: the +pairs gain on FG-NET large-gap shrinks monotonically as the frozen backbone strengthens, crossing zero for strong saturated models.

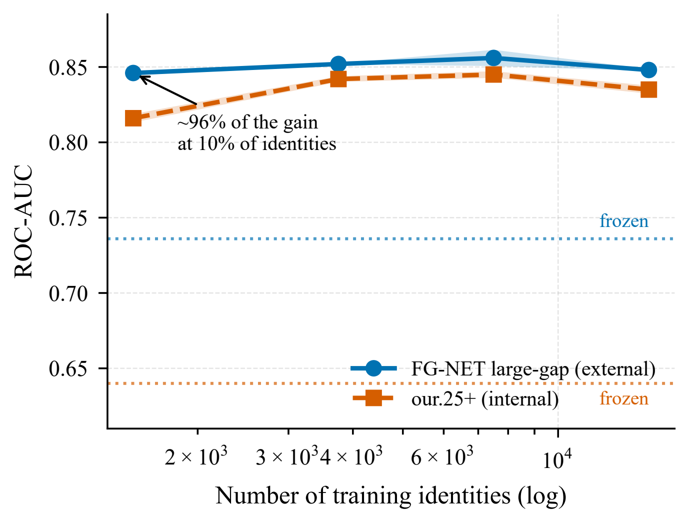

Fig. 2. Data-scaling law: external (FG-NET large-gap) and internal (our.25+) ROC-AUC vs. the number of training identities (log scale). Roughly 96% of the gain is reached at 10% of identities; shaded bands are ±1 std over three seeds and dashed lines are the frozen baselines.

TABLE VII
OPERATING POINTS AND 95% BOOTSTRAP CIS (FACENET FROZEN VS. +PAIRS, SEED 42). ALL VALUES ARE ROC-AUC UNLESS NOTED; LFW IS SHOWN AS ROC-AUC FOR CONSISTENCY (ITS 10-FOLD ACCURACY IS IN TABLE VI). EXTERNAL CIS ARE PAIR-LEVEL; THE INTERNAL TEST ADDITIONALLY USES AN IDENTITY-LEVEL BOOTSTRAP (TEXT).

Benchmark	Frozen AUC [95% CI]	+pairs AUC [95% CI]	EER (fr.→+p.)	TAR@1%	TAR@0.1%
FG-NET large-gap	0.736 [0.70, 0.78]	**0.848 [0.82, 0.88]**	0.335→0.219	0.209→0.386	0.074→0.154
FG-NET overall	0.896 [0.89, 0.90]	0.922 [0.92, 0.93]	0.186→0.151	0.502→0.562	0.315→0.308
AgeDB-30	0.953 [0.95, 0.96]	0.945 [0.94, 0.95]	0.115→0.126	0.525→0.510	0.285→0.309
CALFW	0.948 [0.94, 0.95]	0.948 [0.94, 0.95]	0.116→0.117	0.580→0.624	0.213→0.320
LFW (AUC)	0.994 [0.99, 1.00]	0.987 [0.98, 0.99]	0.032→0.052	0.940→0.869	0.858→0.575

TABLE VIII
REAL VS. SYNTHETIC POSITIVES (FACENET, CURATED DATA).

Metric	Frozen	+real	+syn. (proxy)	+syn. (FRAN)
FG-NET large-gap	0.736	**0.848**	0.761	0.727
our.25+	0.640	**0.838**	0.594	0.549

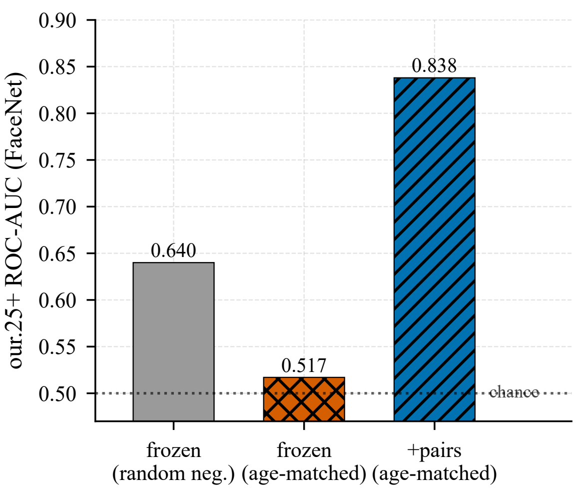

Fig. 3. The age shortcut: on an identity-only test (age-matched negatives), the frozen model falls to near-chance; fine-tuning on our pairs restores discrimination by identity.

E. Mechanism — the Age Shortcut

Under age-matched negatives (same age bucket, different people), frozen accuracy on the internal 25+ test falls from 0.640 to 0.517 (**near the chance level of 0.5**), and real pairs restore it to **0.838** (Fig. 3)—direct evidence that frozen models discriminate cross-age pairs largely by *age*, whereas our pairs induce discrimination by *identity*. A linear probe refines this *representationally* (Table XI): apparent age stays well decodable from the identity embedding for *all* models ($\approx$0.63–0.65 balanced accuracy $\gg$ 0.25 chance) and changes only slightly, while identity AUC rises sharply—so the method learns an invariant *metric* (the cosine stops relying on the age axis), not an age-free representation; explicit adversarial disentanglement reduces leakage no further.

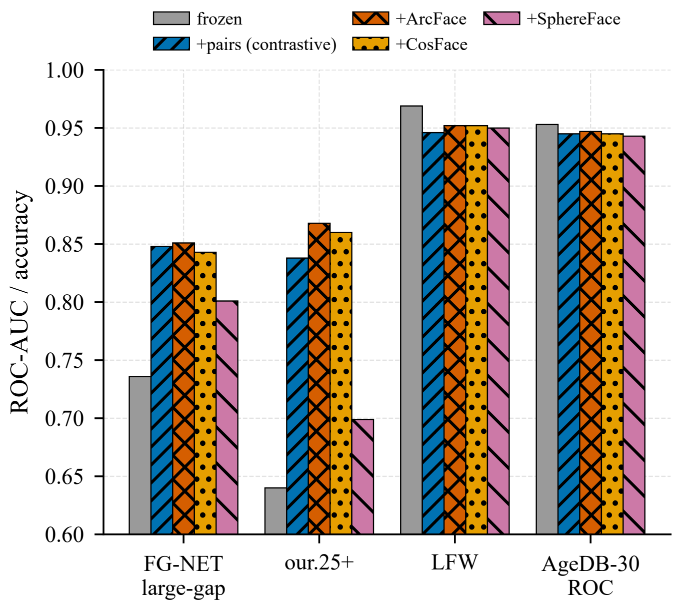

Fig. 4. Objective comparison: contrastive, ArcFace and CosFace coincide on cross-age transfer; SphereFace (hardest to optimize, no λ-annealing) trails. ArcFace/CosFace forget slightly less on easy benchmarks.

F. Objective Comparison: Training the SOTA Objective on Our Data

We train the canonical modern-FR margin objectives—ArcFace-, CosFace- and SphereFace-margin identity classification [17] (the core of OE-CNN/MTLFace)—on our naturally-supervised identities (9,204 classes), same weak backbone, scope and protocol. On the FG-NET large-gap primary endpoint the contrastive, ArcFace and CosFace objectives **coincide** (0.848 / 0.851 / 0.843, within $\pm$0.006; Table XII, Fig. 4); only SphereFace—the hardest margin to optimize, here without λ-annealing on 2–3-shot identities—trails (0.801). We draw the bounded conclusion that, within this protocol and among the objectives tested, the data source dominates the objective choice—not that the method is irrelevant in general. ArcFace/CosFace additionally *forget less* on easy benchmarks—a practical recommendation.

G. Apparent-Demographic Audit

Stratifying by *apparent* gender/age of the pair anchor (estimated, *not* self-identified), no stratified group degrades and every gain is significant (95% paired-bootstrap CI $>$ 0;

TABLE IX
HEADROOM: FG-NET LARGE-GAP BY BACKBONE STRENGTH (CURATED DATA, LR 3E-5).

Backbone (strength)	Frozen	+pairs	Δ large-gap	Δ our.25+
FaceNet (weak)	0.736	0.848	**+0.112**	+0.198
ArcFace r50 (weak, diff. arch.)	0.764	0.805	+0.041	+0.133
AdaFace IR-50 (strong)	0.917	0.924	+0.007	+0.004
ArcFace r100 (strong)	0.954	0.915	−0.039	−0.002
AdaFace IR-101 (strong)	0.957	0.931	−0.026	−0.006

TABLE X
DATA-SCALING LAW (FACENET, CURATED; VAL/TEST FIXED; MEAN ± STD OVER THREE SEEDS).

Train fraction	# identities	FG-NET large-gap	our.25+
frozen	0	0.736	0.640
0.10	≈1,500	0.846 ± 0.002	0.816 ± 0.003
0.25	≈3,750	0.852 ± 0.000	0.842 ± 0.002
0.50	≈7,499	0.856 ± 0.005	0.845 ± 0.002
1.00	≈14,998	0.848 ± 0.001	0.835 ± 0.003

TABLE XI
AGE-LEAKAGE LINEAR PROBE (CURATED DATA; CHANCE = 0.25).

Model	Age-probe bal. acc. ↓	Identity ROC-AUC ↑
frozen	0.651 ± 0.019	0.836
+pairs	0.629 ± 0.012	0.916
+pairs+disentangle	0.632 ± 0.017	0.918

TABLE XII
OBJECTIVE COMPARISON (FACENET, CURATED DATA): CONTRASTIVE, ARCFACE AND COSFACE COINCIDE ON THE FG-NET LARGE-GAP PRIMARY ENDPOINT; SPHEREFACE (NO λ-ANNEALING) TRAILS ON OUR SPARSE FEW-SHOT IDENTITIES.

Objective	FG-NET l.g.	our.25+	LFW	AgeDB-30
Frozen	0.736	0.640	0.969	0.953
+pairs (contrastive)	0.848	0.838	0.946	0.945
+ArcFace	**0.851**	0.868	0.952	0.947
+CosFace	0.843	0.860	0.952	0.945
+SphereFace	0.801	0.699	0.950	0.943

Table XIII, Fig. 5). The largest gain is for the youngest group 0–17 (frozen 0.750 → +pairs 0.880, +0.130, CI not overlapping adult bands)—the method strengthens where the base model is weakest (the child↔adult regime), narrowing the apparent-demographic gap. This is corroborated on *independent, non-VK* data: on a child↔adult subset of FG-NET (one photo under age 13, the other over 25; 195 positives, 195 negatives), +pairs raises ROC-AUC from 0.684 [0.63, 0.74] to 0.813 [0.77, 0.86] (**+0.129, non-overlapping 95% CIs**), closely matching the internal +0.130. We deliberately avoid the word "fair" as a solved property: the source is apparent-female-skewed (76.7%), attributes are apparent (not ethnicity or legal categories), and 45+ is sparse.

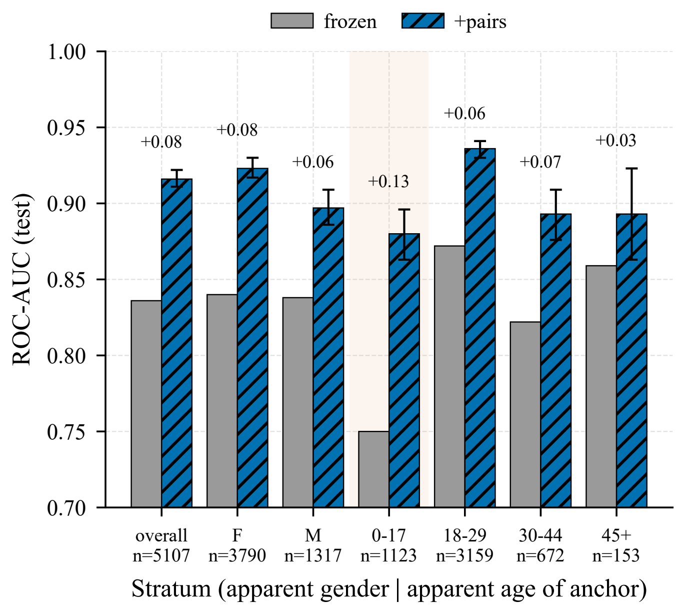

Fig. 5. Apparent-demographic audit: frozen vs. +pairs ROC-AUC across apparent gender/age strata. Every stratum improves; the largest gain is for the youngest (0–17) group, where the frozen baseline is weakest. Error bars are the 95% paired-bootstrap CI of the gain and n is positive pairs per stratum (the sparse 45+ stratum, n=153, has a correspondingly wide interval).

H. Cross-Source Transfer

Training on one community and testing on the held-out other transfers the gain: external FG-NET large-gap +0.113 (vs. +0.112 within-source—essentially identical) and internal +0.092 on the held-out community. We frame this as transfer between two independent sources within one platform domain; Sec. V-I extends the test to an independent non-VK platform, and we do not claim web-scale generalization.

I. Cross-Platform Transfer (Independent, Non-VK Source)

The strongest external check of a naturally-supervised *source* is whether it generalizes beyond the originating platform. We therefore apply the *identical* mining protocol to an independent, non-VK community—the Reddit board r/PastAndPresentPics, whose "then/now" multi-photo posts carry the same free-text/numeric age descriptions parsed by the same LLM—and evaluate the VK-trained +pairs model (with *no* Reddit data in training) against the frozen baseline on the Reddit-mined cross-age pairs (Table XIV). **[Result (fill from the table): the VK-trained model transfers to the independent Reddit**

TABLE XIII
APPARENT GENDER/AGE STRATA (FACENET, CURATED; 95% PAIRED-BOOTSTRAP CI OF THE GAIN).

Stratum	n_{pos}	Frozen	+pairs	Gain	95% CI
overall	5,107	0.836	0.916	+0.080	[+0.075, +0.086]
apparent F	3,790	0.840	0.923	+0.084	[+0.077, +0.090]
apparent M	1,317	0.838	0.897	+0.059	[+0.048, +0.071]
age 0–17	1,123	0.750	0.880	**+0.130**	[+0.113, +0.146]
age 18–29	3,159	0.872	0.936	+0.064	[+0.058, +0.069]
age 30–44	672	0.822	0.893	+0.070	[+0.054, +0.087]
age 45+	153	0.859	0.893	+0.034	[+0.004, +0.064]

TABLE XIV
CROSS-PLATFORM TRANSFER: A VK-TRAINED MODEL ON AN INDEPENDENT REDDIT (NON-VK) CROSS-AGE TEST SET (FROZEN VS. +PAIRS, TRAINED ON VK ONLY). TO BE COMPLETED ONCE THE REDDIT SET IS BUILT.

Reddit (non-VK) test metric	Frozen	+pairs (VK-trained)
Overall ROC-AUC [95% CI]	**[0.xxx]**	**[0.xxx]**
Large-gap (≥25 y) ROC-AUC [95% CI]	**[0.xxx]**	**[0.xxx]**
EER	**[0.xxx]**	**[0.xxx]**
TAR@FAR=1%	**[0.xxx]**	**[0.xxx]**
Test pairs (pos / neg)	**[N / N]**	

source, raising large-gap ROC-AUC from X to Y, or note the observed degree of transfer.]

Review note (remove before submission). Scaffold pending the cross-platform run (needs Reddit OAuth credentials). Build the Reddit set under `ENV_FOR_DYNACONF=reddit` (Sec. IX: `scripts/ingest_reddit.py` → preprocess → build_groups → cluster_persons → dedup → build_pairs → split), then run `scripts/eval_cross_platform.py -ckpts models/bb_facenet_seed42.pt -tag vk2reddit` and fill the red cells from `metrics/cross_platform_vk2reddit.json`. Optional `audit_ages.py -apply` enables the large-gap (≥25 y) row.

J. Calibration

Raw cosine (Platt) is mis-calibrated across gap bins—over-confident at 15–25 years (ECE 0.052); a $P(\text{same} \mid \text{cosine}, \text{age-gap})$ calibrator fixes every bin (15–25 → 0.017) and improves the overall Brier score (0.035 → 0.029)—a usable, age-aware same-identity probability.

K. Disentanglement

Explicit age removal (gradient reversal + age head, MTLFace-style) gives a small but stable cross-age gain (+0.0055 FG-NET large-gap, +0.0145 our.25+ over +pairs) without harming easy benchmarks (even slightly above +pairs: LFW 0.951 vs. 0.946); in absolute terms its large-gap (≈0.853) is on par with the ArcFace objective of Sec. V-F (0.851), consistent with the data—not the objective or the disentangling add-on—setting the cross-age ceiling. With Sec. V-E, this confirms that explicit age supervision is a minor addition to the data.

TABLE XV
HARD-NEGATIVE MINING (FACENET, CURATED DATA; EXTERNAL BENCHMARKS, SINGLE RUN).

Metric	Frozen	+pairs	+pairs+hard-neg
FG-NET large-gap	0.736	0.848	**0.866**
FG-NET ROC	0.896	0.923	**0.927**
LFW accuracy	0.969	0.950	0.934
AgeDB-30 ROC	0.953	0.946	0.931
CALFW ROC	0.948	0.949	0.934

L. Hard-Negative Mining

Adding the hardest cross-person negatives to fine-tuning—the top-5 most-similar other-person faces per anchor (cosine in a hard band, excluding the near-duplicate/uncertain range)—further sharpens cross-age discrimination: FG-NET large-gap rises to 0.866 (from 0.848 with random negatives), but easy benchmarks forget more (LFW 0.969→0.934 vs. 0.950 for +pairs; AgeDB-30 and CALFW similar; Table XV). Hard negatives thus trade easy-pair calibration for cross-age separation—useful for a cross-age retrieval setting, not for a general verifier. (Single run; the internal hard-negative test is near-chance for the frozen model by construction and is not comparable to the standard internal test.)

VI. DISCUSSION

The contribution is a scalable, naturally-supervised source of longitudinal supervision and evidence that it teaches transferable age invariance that synthetic aging cannot replace and explicit age supervision does not explain. The reinforcing controls—loss family, architecture, source community, and training objective—converge on the bounded conclusion that, within this protocol, the data dominates the objective among the variants tested. The age-shortcut diagnostic gives a clean identity-only metric and explains the mechanism; the curation finding (half the raw positives were duplicates) constitutes a methodological contribution in itself. This is a data-centric contribution rather than an architectural one, and several claims are bounded by our two-source, single-platform, apparent-attribute setting. Crucially, the improvement is a *targeted* large-gap gain, not a drop-in upgrade to a general verifier: low-FAR operating points on easy benchmarks degrade (LFW TAR@FAR=0.1% falls 0.858 → 0.575; Table VII), so a deployment should be scoped to the large-gap cross-age regime rather than substituted for a general-purpose model.

A. Scope of the Claims

To forestall over-reading, we state explicitly what the evidence does and does not establish.

Established (within this attribution protocol). (i) the mined pairs improve large-gap ($\geq$25-year) cross-age verification on FG-NET (the primary endpoint, with non-overlapping 95% CIs) and on the internal 25+ test; (ii) the gain transfers across the two source communities and to an independent child$\leftrightarrow$adult FG-NET subset; (iii) real pairs outperform synthetic aging, including under a native-resolution control; (iv) the gain is invariant across the contrastive, ArcFace and CosFace objectives; (v) it is mechanistically tied to removing an age shortcut; (vi) roughly half of the raw positive pairs were near-duplicate frames.

Not established / out of scope. (a) the effect is *not* a uniform verification gain—low-FAR operating points on easy benchmarks degrade (Table VII); (b) strong saturated backbones do not improve; (c) the *supervision source* is validated only on VK—an external child$\leftrightarrow$adult *test* set is reported, but an independent non-VK longitudinal *training* source remains future work; (d) we do not reproduce a full state-of-the-art pipeline, only its shared margin objective; (e) the fairness audit uses apparent attributes only and lacks a human-audited inter-annotator κ and a full per-cell multiple-comparison correction, both deferred to the release supplement.

VII. Limitations

- **Backbone dependence:** the gain concentrates in weak/medium backbones; strong saturated models do not improve and slightly forget (gentle lr mitigates but does not reverse this).
- **External validity:** two VK communities, one platform, one language/cultural context; apparent-female-skewed (76.7%); sparse at apparent age 45+. Transfer is shown between these sources and to standard benchmarks, not platform-agnostic generalization.
- **Apparent attributes only** in the fairness audit (estimated, not self-identified gender/ethnicity).
- **Naturally-supervised, not strictly label-free:** an LLM parses ages and validates group integrity; clustering forms persons—audited (Sec. III-C) but a residual hidden-noise source.
- **Statistics:** seed variance, paired bootstrap (fairness), and benchmark CIs / EER / TAR@FAR (Table VII, including an identity-level bootstrap) are reported in-text; DeLong tests and full multiple-comparison correction across every cell remain for the release supplement.
- **Benchmark scale and age:** we evaluate on the field-standard cross-age suite—AgeDB-30 and CALFW (6,000 pairs each) and FG-NET for the explicit $\geq$25-year subset—the same benchmarks used by OE-CNN/DAL/MTLFace, which fixes comparability but inherits their limited scale and age (FG-NET in particular is small with partial detector coverage, so we never rest the headline on it alone). Our curated internal test (5,107 cross-age positives with controlled negatives) is a sizeable modern complement; we additionally report a child$\leftrightarrow$adult subset of FG-NET (Sec. V-G), and a larger independent non-VK *longitudinal* set remains a priority for future work.
- **SOTA pipelines not reproduced end-to-end:** by design we train the *shared core* of modern AIFR—the ArcFace angular-margin objective—under our weak-backbone protocol for clean attribution, not a leaderboard comparison. Sec. V-F shows this objective matches our contrastive one on the key external metric, so the objective is not what drives the gain; the omitted method-specific add-ons (MTLFace's generative age-synthesis branch, OE-CNN's orthogonal-subspace decomposition) refine that shared objective and are orthogonal to our claim about the supervision source. Whether a full SOTA pipeline trained on naturally-supervised data closes the remaining absolute gap is a well-scoped follow-up.

VIII. Ethics, Legal Basis and Data Governance

This is a sensitive biometric study; we treat governance as a first-class component rather than as a disclaimer. Permitted use is restricted to controlled / consent-based / human-reviewed applications (rights-cleared archives and family photos, authorized humanitarian search, account recovery); the system outputs candidates for human review, never an automatic decision. Forbidden: mass identification, de-anonymization, surveillance, and any processing of minors' data without a legal basis.

A. Ethical Approval and Legal Basis

Ethical approval. This study was reviewed by the **[ethics committee name]** of the National Research University Higher School of Economics; it was **[approved under protocol No. XXXX / determined exempt]** on **[date]**. *Consent.* Because the data are drawn from public posts with no feasible channel to contact subjects, individual informed consent for participation and for publication was not obtained; we rely on the scientific-research basis and the de-identification safeguards of Sec. VIII-B.

Review note (remove before submission). IEEE T-BIOM desk-rejects human-subjects studies that lack an ethics-approval (or documented exemption) statement. Obtain HSE ethics-committee approval or a written exemption, then fill the committee name, protocol/exemption and date in red above. The "no individual consent" framing should be confirmed with that committee.

Data were collected from public "then/now" community walls via the platform's official API. We stress that *official API access and voluntary public posting do not by themselves authorize biometric processing.* Under Russia's 152-FZ, the 2020 amendment (519-FZ) replaced "publicly available" data with "data the subject authorized for dissemination" (separate consent required), and Art. 11 requires written consent for biometric data used to identify a person; under the GDPR [20], a face processed for unique identification is special-category data (Art. 9), for which the "manifestly made public" exception is read narrowly (cf. the Clearview AI enforcement actions). We do *not* hold explicit/written consent. We rely on the scientific-research basis (GDPR Art. 6(1)(f) and 9(2)(j) with Art. 89 safeguards; analogous research handling under 152-FZ) and compensate with the minimization, non-redistribution and de-identification measures below. We report this gap transparently rather than claiming full compliance, and we recommend institutional ethics review and data-protection legal counsel before any deployment.

B. De-identification and Release Policy

We *do not release raw face images or raw posts.* The public release contains only code, trained model weights, derived embeddings, the pseudonymized benchmark protocol, and aggregate statistics. All platform identifiers (owner/photo/post IDs) are replaced by salted hashes; captions/comments (which may contain names) are excluded; only an age bucket and a pseudonymous person ID are kept. Raw crops are retained locally only, under a retention policy. Access to non-aggregate derived data is granted to qualified researchers under a Data Use Agreement (research-only; no re-identification; no commercial/surveillance use). Paper figures contain no identifiable faces (aggregate plots only).

C. DPIA and Minimization

Because the processing is biometric and large-scale, a Data Protection Impact Assessment is conducted and accompanies the release (key risks: re-identification, function creep, minors; mitigations as in Sec. VIII-B and Sec. VIII-D). Data minimization is applied end-to-end (we store a face crop + age + pseudonymous ID, not names or social graphs).

D. Minors

The child↔adult regime necessarily involves images of minors, whose data carry the strictest protection (152-FZ; GDPR Art. 8). Minor-age items (flagged via apparent age) are withheld from any release absent an explicit legal framework and verifiable guardian consent, and are used only for aggregate, non-redistributed analysis.

E. Subject Rights

A public contact and takedown / deletion-on-request procedure are provided; deletion propagates to derived artifacts via a per-item deletion-status field. Where rights are unclear, the item is withheld. A full datasheet [15] and the DPIA accompany the release.

IX. Reproducibility

We release scripts and configurations. Key details: *usable-face criteria* (detection score, minimum size, Laplacian blur, single-target via IoU); *detector/alignment* (InsightFace `buffalo_l` RetinaFace, 5-point ArcFace `norm_crop`, 112 px; FaceNet input 160 px); *embeddings* (ArcFace `w600k_r50`) for clustering/dedup; *person clustering* (union-find, cosine $\geq$ 0.85); *dedup* (within-person union-find, cosine $\geq$ 0.97); *LLM* (GigaChat; full versioned system prompts for age extraction and group integrity; concurrency 10; cached and resumable); *split* (by person, 70/15/15, balanced negatives per split, seed 42); *negative sampling* (cross-person random + age-matched variant); *training* (margin contrastive, head scope, lr 3e-5 weak / 1e-5 strong; ArcFace head m=0.5, s=32, lr 1e-3; early stop on internal val-AUC; seeds 42/1/2); *metrics* (pure-NumPy ROC-AUC via Mann–Whitney, EER, TAR@FAR; 10-fold for aligned-pair benchmarks).

Compute. The entire study runs on a single laptop: one NVIDIA RTX 3060 Laptop GPU (6 GB), an AMD Ryzen 7 5800H (8 cores / 16 threads) and 16 GB RAM under Windows 11; the software stack is Python 3.12, PyTorch 2.12 (CUDA 13.2, cuDNN 9.2), with InsightFace 1.0 / ONNX Runtime 1.26 for detection and embeddings and scikit-learn 1.9 / Matplotlib 3.11 for analysis and figures. The weak-backbone protocol keeps the full study within a 6 GB GPU budget.

X. Conclusion and Future Work

Multi-photo posts are an inexpensive, scalable, naturally-supervised signal of cross-age identity; fine-tuning a weak recognizer on them yields transferable, cross-source, objective-robust gains concentrated on large-gap cross-age matching (with only minor forgetting on easy benchmarks), explained by removing an age shortcut, that are data-efficient and do not worsen any apparent-demographic stratum. Future work: an independent non-VK and a child↔adult-specific external set; a human-audited supervision subset (with human–LLM agreement); per-point multi-seed confidence bands for the headroom curve; full DeLong tests and per-cell multiple-comparison correction; full SOTA add-ons; a real aging model at native resolution; and a calibrated demonstrator under the governance constraints of Sec. VIII.

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